

1. The ability to do work is called:

- (A) Power
- **(B) Energy**
- (C) Force
- (D) Friction

2. The SI unit of energy is:

- (A) Newton
- (B) Watt
- **(C) Joule**
- (D) Pascal

3. Energy stored in a body due to its position is:

- (A) Kinetic Energy
- **(B) Potential Energy**
- (C) Thermal Energy
- (D) Chemical Energy

4. Energy possessed by a body due to its motion is:

- **(A) Kinetic Energy**
- (B) Potential Energy
- (C) Nuclear Energy
- (D) Solar Energy

5. A stretched rubber band possesses:

- (A) Kinetic Energy
- **(B) Elastic Potential Energy**
- (C) Electrical Energy
- (D) Sound Energy

6. The formula for Kinetic Energy is:

- (A) mgh
- **(B) $\frac{1}{2}mv^2$**
- (C) $F \times d$
- (D) $P \times t$

7. The formula for Gravitational Potential Energy is:

- **(A) mgh**
- (B) mc^2
- (C) $\frac{1}{2}mv^2$
- (D) $V \times I$

8. **Einstein's mass-energy equation is:**

- (A) $E = mgh$
- (B) $E = \frac{1}{2}mv^2$
- **(C) $E = mc^2$**
- (D) $P = VI$

9. **Food possesses which type of energy?**

- (A) Heat
- (B) Light
- **(C) Chemical**
- (D) Nuclear

10. **The ultimate source of energy for Earth is:**

- (A) Coal
- **(B) Sun**
- (C) Moon
- (D) Electricity

11. **Which of the following is a non-renewable source of energy?**

- (A) Wind
- **(B) Natural Gas**
- (C) Sunlight
- (D) Biomass

12. **Coal, oil, and gas are collectively called:**

- (A) Biogas
- **(B) Fossil Fuels**
- (C) Minerals
- (D) Renewables

13. **The main component of natural gas in Pakistan is:**

- (A) Ethane
- **(B) Methane**
- (C) Propane
- (D) Butane

14. **Petroleum is also known as:**

- (A) Black gold
- **(B) Liquid gold**
- (C) Crude oil
- (D) Both B and C

15. **Burning fossil fuels releases which gas responsible for the greenhouse effect?**
- (A) Oxygen
 - (B) Nitrogen
 - **(C) Carbon dioxide**
 - (D) Helium
16. **Large-scale coal deposits in Pakistan are found in:**
- (A) Quetta
 - **(B) Thar**
 - (C) Khewra
 - (D) Sialkot
17. **Chemical energy is converted into heat energy during:**
- (A) Melting
 - (B) Freezing
 - **(C) Combustion**
 - (D) Evaporation
18. **The energy obtained from the interior of the Earth is:**
- **(A) Geothermal Energy**
 - (B) Solar Energy
 - (C) Tidal Energy
 - (D) Wind Energy
19. **Crude oil is refined through:**
- (A) Filtration
 - **(B) Fractional Distillation**
 - (C) Evaporation
 - (D) Oxidation
20. **Which is the purest form of coal?**
- **(A) Anthracite**
 - (B) Bituminous
 - (C) Lignite
 - (D) Peat
21. **Hydroelectric power is generated using:**
- (A) Steam
 - **(B) Falling water**
 - (C) Wind
 - (D) Sunlight

22. **The device that converts solar energy directly into electricity is:**
- (A) Solar Cooker
 - (B) Solar Heater
 - **(C) Solar Cell (Photovoltaic)**
 - (D) Generator
23. **Energy from organic matter (plants/waste) is called:**
- (A) Geothermal
 - **(B) Biomass**
 - (C) Nuclear
 - (D) Hydro
24. **Windmills are used to convert wind energy into:**
- **(A) Electrical energy**
 - (B) Chemical energy
 - (C) Nuclear energy
 - (D) Heat energy
25. **The energy produced by the movement of ocean waves is:**
- (A) Nuclear
 - (B) Geothermal
 - **(C) Tidal energy**
 - (D) Solar
26. **Which of these is a clean source of energy?**
- (A) Coal
 - (B) Oil
 - **(C) Wind**
 - (D) Wood
27. **Biogas is primarily a mixture of:**
- (A) Hydrogen and Oxygen
 - **(B) Methane and Carbon dioxide**
 - (C) Nitrogen and Helium
 - (D) Butane and Propane
28. **Hydel power stations are mostly built near:**
- (A) Deserts
 - **(B) Rivers and Dams**
 - (C) Oceans
 - (D) Forests
29. **Solar panels are usually made of:**

- (A) Iron
 - (B) Copper
 - **(C) Silicon**
 - (D) Aluminum
30. **Which energy source does not depend on the Sun?**
- (A) Wind
 - (B) Biomass
 - **(C) Geothermal**
 - (D) Hydel
31. **Nuclear energy is released during:**
- (A) Combustion
 - **(B) Nuclear Fission**
 - (C) Oxidation
 - (D) Evaporation
32. **The fuel used in nuclear power plants is usually:**
- (A) Carbon
 - **(B) Uranium**
 - (C) Hydrogen
 - (D) Nitrogen
33. **In nuclear fission, a heavy nucleus:**
- **(A) Splits into lighter nuclei**
 - (B) Combines with another nucleus
 - (C) Remains the same
 - (D) Turns into gas
34. **Nuclear fusion occurs naturally in:**
- (A) Volcanoes
 - (B) Oceans
 - **(C) Stars and the Sun**
 - (D) Earth's core
35. **The KANUPP plant in Pakistan produces energy using:**
- (A) Coal
 - (B) Gas
 - **(C) Nuclear fuel**
 - (D) Water
36. **The safety hazard associated with nuclear energy is:**
- (A) Smoke
 - **(B) Radiation**

- (C) Acid rain
- (D) Deforestation
- 37. **Control rods in a nuclear reactor are made of:**
 - **(A) Boron or Cadmium**
 - (B) Iron
 - (C) Lead
 - (D) Copper
- 38. **Fission releases energy because:**
 - **(A) Mass is converted into energy**
 - (B) Heat is added
 - (C) Protons disappear
 - (D) Electrons move faster
- 39. **The first nuclear power plant in Pakistan was:**
 - (A) CHASNUPP
 - **(B) KANUPP**
 - (C) Hubco
 - (D) Kapco
- 40. **Heavy water is often used in reactors as a:**
 - (A) Fuel
 - **(B) Moderator**
 - (C) Casing
 - (D) Exhaust
- 41. **Energy can neither be created nor destroyed is the Law of:**
 - (A) Motion
 - (B) Gravity
 - **(C) Conservation of Energy**
 - (D) Thermodynamics
- 42. **In an electric bulb, electrical energy changes into:**
 - **(A) Light and Heat energy**
 - (B) Sound energy
 - (C) Chemical energy
 - (D) Magnetic energy
- 43. **A battery converts chemical energy into:**
 - (A) Mechanical energy
 - **(B) Electrical energy**
 - (C) Nuclear energy
 - (D) Light energy

44. **In a microphone, sound energy is converted into:**
- **(A) Electrical energy**
 - (B) Magnetic energy
 - (C) Heat energy
 - (D) Kinetic energy
45. **When we rub our hands, kinetic energy changes to:**
- (A) Light energy
 - **(B) Heat energy**
 - (C) Sound energy
 - (D) Chemical energy
46. **In a dam, the potential energy of water is first converted into:**
- **(A) Kinetic energy**
 - (B) Electrical energy
 - (C) Heat energy
 - (D) Sound energy
47. **A generator converts mechanical energy into:**
- (A) Chemical energy
 - **(B) Electrical energy**
 - (C) Nuclear energy
 - (D) Solar energy
48. **An electric motor converts electrical energy into:**
- **(A) Mechanical energy**
 - (B) Heat energy
 - (C) Chemical energy
 - (D) Light energy
49. **In photosynthesis, plants convert solar energy into:**
- (A) Kinetic energy
 - **(B) Chemical energy**
 - (C) Electrical energy
 - (D) Nuclear energy
50. **Total energy in a closed system remains:**
- (A) Increasing
 - (B) Decreasing
 - **(C) Constant**
 - (D) Zero
51. **The ratio of useful output energy to total input energy is:**

- (A) Power
 - **(B) Efficiency**
 - (C) Friction
 - (D) Force
52. **Efficiency is usually expressed in:**
- (A) Joules
 - (B) Watts
 - **(C) Percentage**
 - (D) Newtons
53. **The rate of doing work is called:**
- **(A) Power**
 - (B) Energy
 - (C) Momentum
 - (D) Torque
54. **The SI unit of power is:**
- (A) Joule
 - **(B) Watt**
 - (C) Ampere
 - (D) Volt
55. **One Horsepower (hp) is equal to:**
- (A) 500 Watts
 - **(B) 746 Watts**
 - (C) 1000 Watts
 - (D) 786 Watts
56. **Formula for Power is:**
- **(A) W/t**
 - (B) $F \times d$
 - (C) mgh
 - (D) mc^2
57. **If a machine does 100J of work in 10s, its power is:**
- (A) 1000W
 - **(B) 10W**
 - (C) 110W
 - (D) 90W
58. **Energy dissipated as heat in machines is usually due to:**
- (A) Gravity
 - **(B) Friction**

- (C) Magnetism
- (D) Inertia
- 59. **A system with 100% efficiency is:**
 - (A) Common
 - **(B) Impossible**
 - (C) Only in solar cells
 - (D) Only in nuclear plants
- 60. **Kilowatt-hour (kWh) is a unit of:**
 - (A) Power
 - **(B) Energy**
 - (C) Time
 - (D) Force
- 61. **Acid rain is caused by oxides of:**
 - (A) Carbon and Hydrogen
 - **(B) Sulfur and Nitrogen**
 - (C) Helium and Neon
 - (D) Oxygen and Argon
- 62. **Global warming is primarily caused by the increase of:**
 - (A) Oxygen
 - **(B) Carbon dioxide**
 - (C) Nitrogen
 - (D) Ozone
- 63. **The thinning of the ozone layer is caused by:**
 - (A) CO₂
 - **(B) Chlorofluorocarbons (CFCs)**
 - (C) Methane
 - (D) Water vapor
- 64. **Which energy source is most harmful to the environment?**
 - **(A) Coal**
 - (B) Solar
 - (C) Wind
 - (D) Hydroelectric
- 65. **To conserve energy at home, we should use:**
 - (A) Incandescent bulbs
 - **(B) LED bulbs**
 - (C) High power heaters
 - (D) Old appliances

66. **The term "procurement" in energy refers to:**
- (A) Wasting energy
 - **(B) Obtaining/Acquiring energy**
 - (C) Burning energy
 - (D) Destroying energy
67. **Solar energy is available only during:**
- (A) Night
 - **(B) Day**
 - (C) Winter
 - (D) Rainy weather
68. **Which country is a leader in wind energy production?**
- (A) Pakistan
 - (B) Saudi Arabia
 - **(C) Denmark**
 - (D) Kuwait
69. **Deforestation leads to a decrease in:**
- (A) CO₂
 - **(B) Oxygen**
 - (C) Soil erosion
 - (D) Heat
70. **Pollution caused by heat release into water bodies is:**
- (A) Air pollution
 - (B) Noise pollution
 - **(C) Thermal pollution**
 - (D) Soil pollution
71. **Tidal energy is produced due to the gravitational pull of:**
- (A) Sun
 - **(B) Moon**
 - (C) Mars
 - (D) Venus
72. **The temperature at the center of the Sun is about:**
- (A) 1 million K
 - **(B) 15 million K**
 - (C) 1000 K
 - (D) 5000 K
73. **Energy in the form of electromagnetic waves is:**
- (A) Sound

- **(B) Radiation (Light)**
 - (C) Chemical
 - (D) Potential
74. **In a thermocouple, heat is converted into:**
- **(A) Electricity**
 - (B) Sound
 - (C) Motion
 - (D) Magnetism
75. **The energy of moving air is:**
- (A) Solar
 - (B) Chemical
 - **(C) Wind energy**
 - (D) Tidal
76. **Which of the following has the highest kinetic energy?**
- (A) A parked car
 - **(B) A flying bullet**
 - (C) A book on a table
 - (D) A sleeping person
77. **Which form of energy is easiest to transport?**
- (A) Heat
 - **(B) Electricity**
 - (C) Sound
 - (D) Mechanical
78. **Geothermal plants use _____ to turn turbines.**
- (A) Water
 - **(B) Steam from underground**
 - (C) Wind
 - (D) Natural gas
79. **Energy stored in the nucleus of an atom is:**
- (A) Chemical
 - **(B) Nuclear**
 - (C) Electrical
 - (D) Thermal
80. **An example of biomass is:**
- (A) Iron ore
 - (B) Sand
 - **(C) Sugarcane waste**

- (D) Plastic
- 81. **Sound is a form of _____ energy.**
 - (A) Chemical
 - (B) Potential
 - **(C) Mechanical (vibrational)**
 - (D) Nuclear
- 82. **Work is said to be done when:**
 - (A) A force acts
 - **(B) A force moves an object**
 - (C) Energy is stored
 - (D) An object is at rest
- 83. **If velocity of a body is doubled, its K.E becomes:**
 - (A) Double
 - (B) Half
 - **(C) Four times**
 - (D) Same
- 84. **Which fossil fuel is the cleanest burning?**
 - (A) Coal
 - (B) Oil
 - **(C) Natural Gas**
 - (D) Wood
- 85. **Sui is famous in Pakistan for:**
 - (A) Coal
 - (B) Oil
 - **(C) Natural Gas**
 - (D) Solar power
- 86. **In an electric iron, electrical energy changes to:**
 - (A) Light
 - **(B) Heat**
 - (C) Sound
 - (D) Chemical
- 87. **The major source of electricity in Pakistan is:**
 - (A) Nuclear
 - (B) Solar
 - **(C) Thermal (and Hydel)**
 - (D) Wind
- 88. **A loudspeaker converts electrical energy into:**

- (A) Light
 - **(B) Sound**
 - (C) Heat
 - (D) Chemical
- 89. **Burning of wood is a _____ change.**
 - (A) Physical
 - **(B) Chemical**
 - (C) Nuclear
 - (D) Biological
- 90. **One Joule per second is equal to:**
 - (A) 1 Newton
 - **(B) 1 Watt**
 - (C) 1 Pascal
 - (D) 1 Volt
- 91. **Fossil fuels are found in:**
 - **(A) Sedimentary rocks**
 - (B) Igneous rocks
 - (C) Metamorphic rocks
 - (D) Volcanic craters
- 92. **Solar energy can be stored in:**
 - (A) Transformers
 - **(B) Batteries**
 - (C) Generators
 - (D) Turbines
- 93. **Which is NOT a primary source of energy?**
 - (A) Sun
 - (B) Uranium
 - (C) Wind
 - **(D) Electricity**
- 94. **The efficiency of a typical petrol engine is about:**
 - (A) 90%
 - **(B) 25-30%**
 - (C) 100%
 - (D) 5%
- 95. **Greenhouse effect leads to:**
 - (A) Cooling of Earth
 - **(B) Warming of Earth**

- (C) No change
 - (D) Increase in Oxygen
96. **In a pendulum, P.E is maximum at:**
- (A) Mean position
 - **(B) Extreme positions**
 - (C) Bottom
 - (D) All points
97. **Energy from the Sun reaches Earth through:**
- (A) Conduction
 - (B) Convection
 - **(C) Radiation**
 - (D) Transmission
98. **Natural gas is mostly used in Pakistan for:**
- (A) Space travel
 - **(B) Cooking and Power generation**
 - (C) Making gold
 - (D) Cooling
99. **Wastes from living things are called:**
- (A) Minerals
 - **(B) Biomass**
 - (C) Fossils
 - (D) Isotopes
100. **Saving energy is also known as: * (A) Energy Conservation**
 * (B) Energy Production * (C) Energy Friction * (D) Energy Procurement